

## SAFETY DATA SHEET

Creation Date 11-Nov-2010

Revision Date 21-May-2025

Revision Number 2

### 1. Identification

**Product Name** Chloroacetyl chloride

**Cat No. :** S15846

**CAS No** 79-04-9  
**Synonyms** Chloroacetic acid chloride.; Chloroacetyl chloride

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                                            |              |
|----------------------------------------------------------------------------|--------------|
| Acute oral toxicity                                                        | Category 3   |
| Acute dermal toxicity                                                      | Category 3   |
| Acute Inhalation Toxicity - Vapors                                         | Category 3   |
| Skin Corrosion/Irritation                                                  | Category 1 A |
| Serious Eye Damage/Eye Irritation                                          | Category 1   |
| Specific target organ toxicity - (repeated exposure)                       | Category 1   |
| Target Organs - Central nervous system (CNS), Gastrointestinal tract (GI). |              |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Causes severe skin burns and eye damage  
Causes damage to organs through prolonged or repeated exposure  
Toxic if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Use only outdoors or in a well-ventilated area  
Do not breathe dust/fume/gas/mist/vapors/spray  
Wear respiratory protection

**Response**

Immediately call a POISON CENTER or doctor/physician

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Immediately call a POISON CENTER or doctor/physician

**Skin**

Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Ingestion**

Rinse mouth  
Do NOT induce vomiting

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Very toxic to aquatic life with long lasting effects  
Reacts violently with water  
Contact with water liberates toxic gas  
Corrosive to the respiratory tract

### 3. Composition/information on Ingredients

| Component             | CAS No  | Weight % |
|-----------------------|---------|----------|
| Chloroacetyl chloride | 79-04-9 | <=100    |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical

|                                            |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | advice.                                                                                                                                                                                                                                                                                                                        |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Inhalation</b>                          | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                          |

## 5. Fire-fighting measures

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                          |
| <b>Flash Point</b>                      | No information available                                          |
| <b>Method -</b>                         | No information available                                          |
| <b>Autoignition Temperature</b>         | No information available                                          |
| <b>Explosion Limits</b>                 |                                                                   |
| <b>Upper</b>                            | No data available                                                 |
| <b>Lower</b>                            | No data available                                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                          |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Contact with water liberates toxic gas. Reacts violently with water. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Thermal decomposition can lead to release of irritating gases and vapors. Phosgene. Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
4

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
W

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.                                                                               |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water.                                                                                                                      |

## 7. Handling and Storage

### Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Handle under an inert atmosphere.

### Storage.

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep away from water or moist air. Do not store in metal containers. Store under an inert atmosphere. Protect from moisture. Incompatible Materials. Alcohols. Bases. Amines. Metals. Water.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component             | ACGIH TLV                               | OSHA PEL                                                        | NIOSH                                                                      | Mexico OEL (TWA)                |
|-----------------------|-----------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------|
| Chloroacetyl chloride | TWA: 0.05 ppm<br>STEL: 0.15 ppm<br>Skin | (Vacated) TWA: 0.05 ppm<br>(Vacated) TWA: 0.2 mg/m <sup>3</sup> | IDLH: 1.3 ppm<br>REL = 0.05 ppm (TWA)<br>REL = 0.2 mg/m <sup>3</sup> (TWA) | TWA: 0.05 ppm<br>STEL: 0.15 ppm |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.  
Ensure adequate ventilation, especially in confined areas.

### Personal Protective Equipment

#### Eye/face Protection

Tight sealing safety goggles. Face protection shield.

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

#### Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

#### Recommended Filter type:

Particulates filter conforming to EN 143. Acid gases filter. Type E. Yellow. conforming to EN14387.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                            |
|----------------------------------|----------------------------|
| Physical State                   | Liquid                     |
| Appearance                       | Clear                      |
| Odor                             | pungent                    |
| Odor Threshold                   | No information available   |
| pH                               | No information available   |
| Melting Point/Range              | -22 °C / -7.6 °F           |
| Boiling Point/Range              | 105 °C / 221 °F @ 760 mmHg |
| Flash Point                      | No information available   |
| Evaporation Rate                 | No information available   |
| Flammability (solid,gas)         | Not applicable             |
| Flammability or explosive limits |                            |
| Upper                            | No data available          |
| Lower                            | No data available          |

|                                        |                             |
|----------------------------------------|-----------------------------|
| Vapor Pressure                         | No information available    |
| Vapor Density                          | No information available    |
| Specific Gravity                       | 1.420                       |
| Solubility                             | Reacts violently with water |
| Partition coefficient; n-octanol/water | No data available           |
| Autoignition Temperature               | No information available    |
| Decomposition Temperature              | No information available    |
| Viscosity                              | No information available    |
| Molecular Formula                      | C2 H2 Cl2 O                 |
| Molecular Weight                       | 112.94                      |

## 10. Stability and reactivity

|                                  |                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                                                                |
| Stability                        | Stable under normal conditions.                                                                                                                                    |
| Conditions to Avoid              | Incompatible products. Excess heat. Exposure to moist air or water. Exposure to moisture.                                                                          |
| Incompatible Materials           | Alcohols, Bases, Amines, Metals, Water                                                                                                                             |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Thermal decomposition can lead to release of irritating gases and vapors, Phosgene, Hydrogen chloride gas |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                                           |
| Hazardous Reactions              | None under normal processing. Reacts violently with water.                                                                                                         |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Oral LD50

ATE = 50 - 300 mg/kg.

##### Component Information

| Component             | LD50 Oral                | LD50 Dermal                                                   | LC50 Inhalation                                           |
|-----------------------|--------------------------|---------------------------------------------------------------|-----------------------------------------------------------|
| Chloroacetyl chloride | LD50 = 200 mg/kg ( Rat ) | LD50 = 662 mg/kg ( Rat )<br>LD50 = 316 - 501 mg/kg ( Rabbit ) | LC50 = 1000 ppm ( Rat ) 4h<br>LC50 = 4.69 mg/L ( Rat ) 4h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | CAUSES (SEVERE) EYE BURNS Causes skin burns                                              |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component             | CAS No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------|---------|------------|------------|------------|------------|------------|
| Chloroacetyl chloride | 79-04-9 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** Central nervous system (CNS) Gastrointestinal tract (GI)

|                                                   |                                                                                                                                                                                                                                                            |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                   |
| <b>Symptoms / effects, both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                                                                                                   |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                             |

## 12. Ecological information

### Ecotoxicity

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. Reacts with water so no ecotoxicity data for the substance is available.

|                                      |                                          |
|--------------------------------------|------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely                  |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                |
| <b>Mobility</b>                      | Is not likely mobile in the environment. |

| Component             | log Pow |
|-----------------------|---------|
| Chloroacetyl chloride | -0.22   |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                                |                       |
|--------------------------------|-----------------------|
| <b>UN-No</b>                   | UN1752                |
| <b>Proper Shipping Name</b>    | CHLOROACETYL CHLORIDE |
| <b>Hazard Class</b>            | 6.1                   |
| <b>Subsidiary Hazard Class</b> | 8                     |
| <b>Packing Group</b>           | I                     |

### TDG

Forbidden  
FORBIDDEN FOR IATA TRANSPORT

### IATA

### IMDG/IMO

|                                |                       |
|--------------------------------|-----------------------|
| <b>UN-No</b>                   | UN1752                |
| <b>Proper Shipping Name</b>    | CHLOROACETYL CHLORIDE |
| <b>Hazard Class</b>            | 6.1                   |
| <b>Subsidiary Hazard Class</b> | 8                     |
| <b>Packing Group</b>           | I                     |

## 15. Regulatory Information

### United States of America Inventory

| Component             | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-----------------------|---------|------|-----------------------------------------------|-----------------------------|
| Chloroacetyl chloride | 79-04-9 | X    | ACTIVE                                        | -                           |

### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b) - Notices of Export**

Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component             | CAS No  | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-----------------------|---------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Chloroacetyl chloride | 79-04-9 | X   | -   | 201-171-6 | X     | X    | X    | X    | X     | KE-05500 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

Not applicable

**Clean Air Act**

Not applicable

**OSHA - Occupational Safety and Health Administration**

Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component             | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-----------------------|---------------|------------|--------------|----------|--------------|
| Chloroacetyl chloride | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
 DOT Marine Pollutant N  
 DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product contains the following DHS chemicals:

**Legend** - STQs = Screening Threshold Quantities, APA = A placarded amount

| Component             | DHS Chemical Facility Anti-Terrorism Standard |
|-----------------------|-----------------------------------------------|
| Chloroacetyl chloride | APA                                           |

**Other International Regulations**

Mexico - Grade

No information available

## Authorisation/Restrictions according to EU REACH

| Component             | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Chloroacetyl chloride | 79-04-9 | -                                                                   | Use restricted. See entry 75.<br>(see link for restriction details)           | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component             | CAS No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| Chloroacetyl chloride | 79-04-9 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component             | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Chloroacetyl chloride | 79-04-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other Information

## Prepared By

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

## Creation Date

11-Nov-2010

## Revision Date

21-May-2025

## Print Date

21-May-2025

## Revision Summary

Initial Release.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**